

M E T A P Y T O R C H O P E N E N V H A C K A T H O N · 2 0 2 6

AUTO - EXAMINER

S E L F - I M P R O V I N G A I T H R O U G H A U T O - C U R R I C U L U M

An RL environment where the agent writes its own coding challenges,
solves them, and the difficulty adapts to its performance.

T E A M V A M P ·

T U S H A R A ·

P A D M A S H R E E ·

P R A N A V

THE PROBLEM

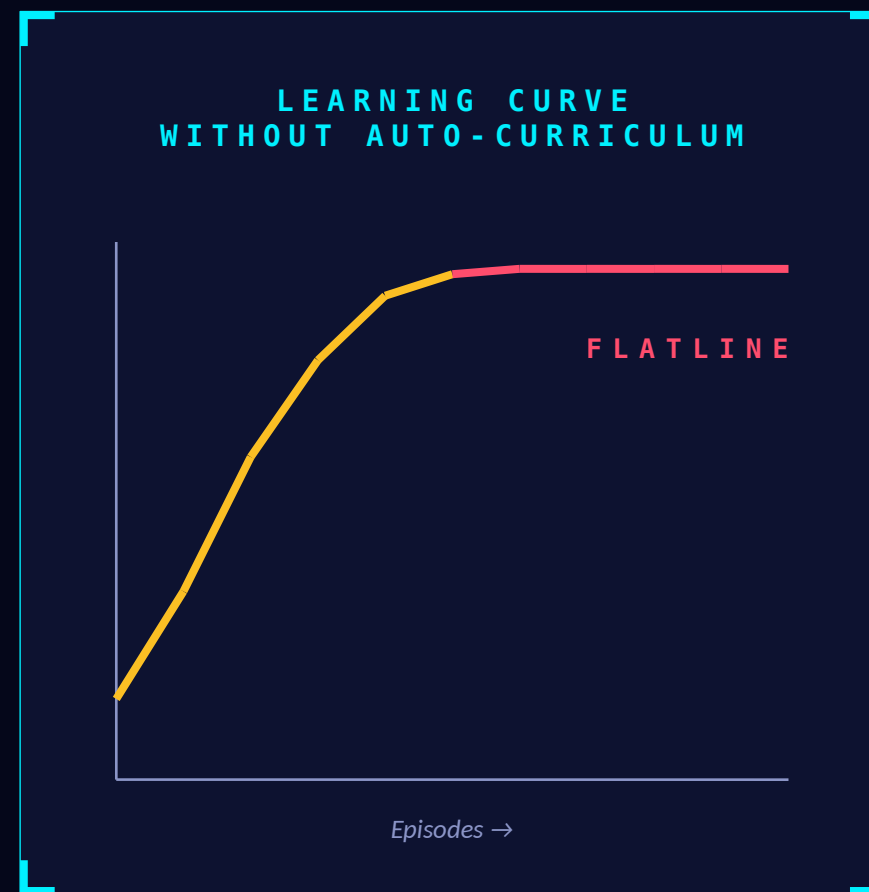
Why fixed environments stall LLM training

RL environments today have a fixed task bank.

The agent grinds through it, masters the tasks, and then the reward curve flatlines.

PAIN POINTS

- ▶ Manual difficulty curation doesn't scale
- ▶ Learning plateaus after initial mastery
- ▶ Static benchmarks teach static skills
- ▶ Adaptive curricula are an unsolved RL problem



OUR SOLUTION

The agent writes the curriculum, not just the answers

01

AGENT GENERATES

Instead of solving fixed tasks, the agent first writes its own coding challenge at the current difficulty.

02

AGENT SOLVES

It then submits a Python solution. The environment auto-generates 5 pytest assertions and grades the solution in a sandboxed subprocess.

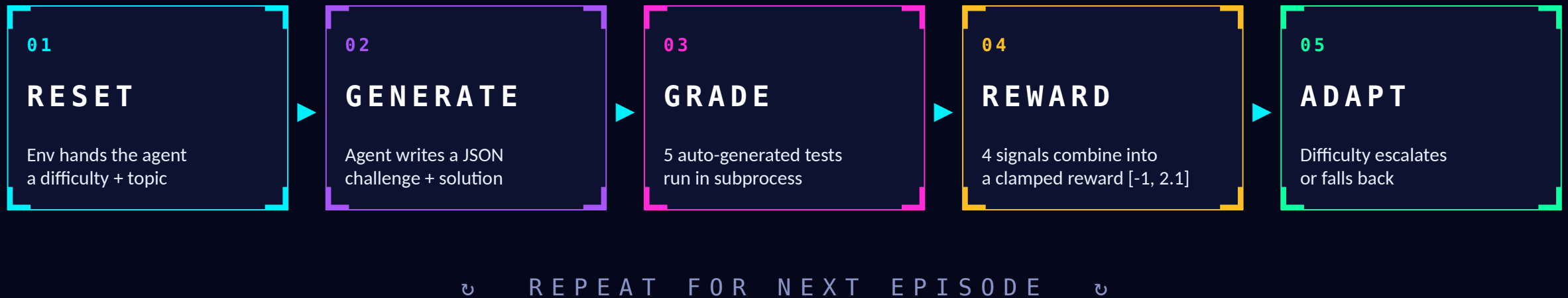
03

DIFFICULTY ADAPTS

Score ≥ 0.8 raises difficulty, < 0.5 lowers it. The curriculum builds itself, episode by episode.

HOW IT WORKS

One episode, end to end



ACTION SHAPE

```
{
  "challenge": "Write a function that returns the nth Fibonacci number...",
  "solution": "def fib(n):\n    if n < 2: return n\n    ..."
```

REWARD FUNCTION

Four independent signals — hard to game

CORRECTNESS

[0.0 → 1.0]

tests_passed / total_tests

DIFFICULTY MULTIPLIER

[× 1.2 → × 2.0]

Harder problems pay more · 1 + level/5

FORMAT COMPLIANCE

[-0.2 → +0.1]

Rewards valid JSON, penalizes empty

TIMEOUT PENALTY

[-0.3 → 0.0]

Punishes infinite loops + crashes

FINAL = CORRECTNESS × DIFFICULTY + FORMAT + TIMEOUT · CLAMPED [-1.0, 2.10]

LIVE DASHBOARD

Watch the agent run in real time



Difficulty path · live reward bars · color-coded scoring · session-persistent terminal logs

RESULTS · EVALUATION

100 episodes · Qwen 2.5 72B

AVG REWARD

1.86

/ 2.10 max

MAX DIFFICULTY

5

/ 5

PERFECT SCORES

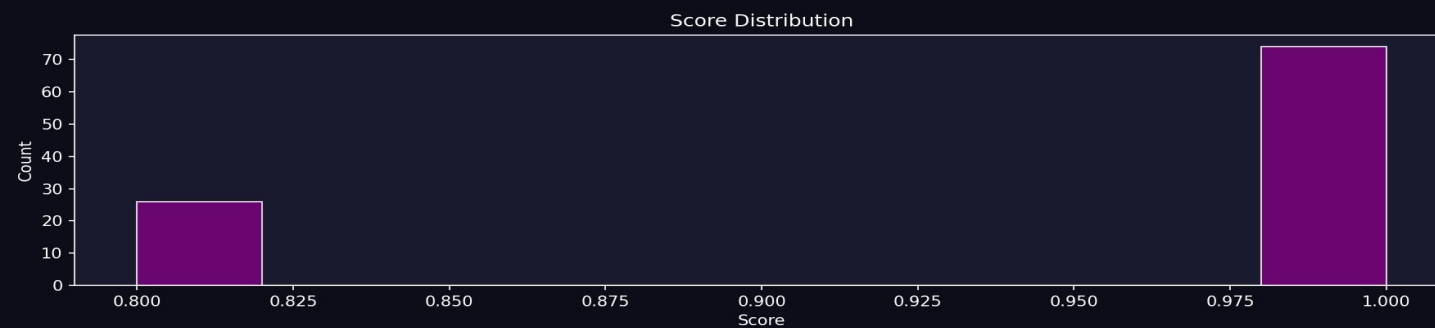
74%

score = 1.0

COMPLETION

100%

0 crashes

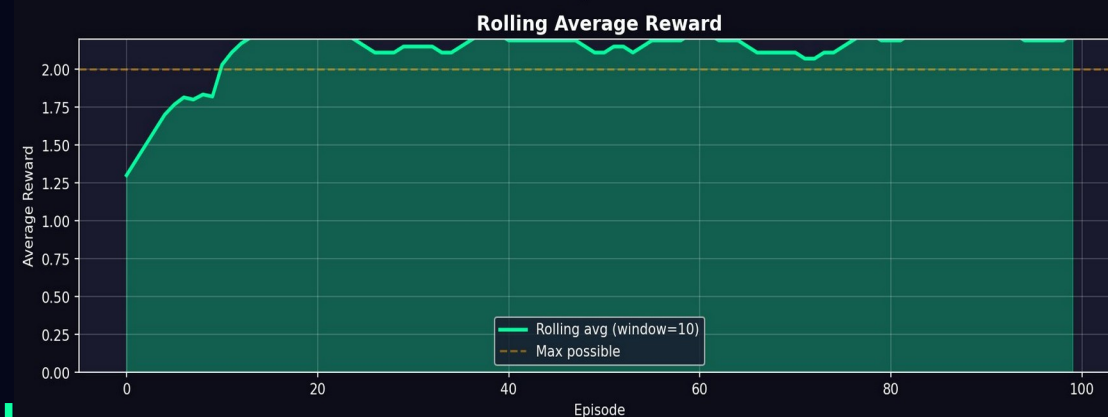
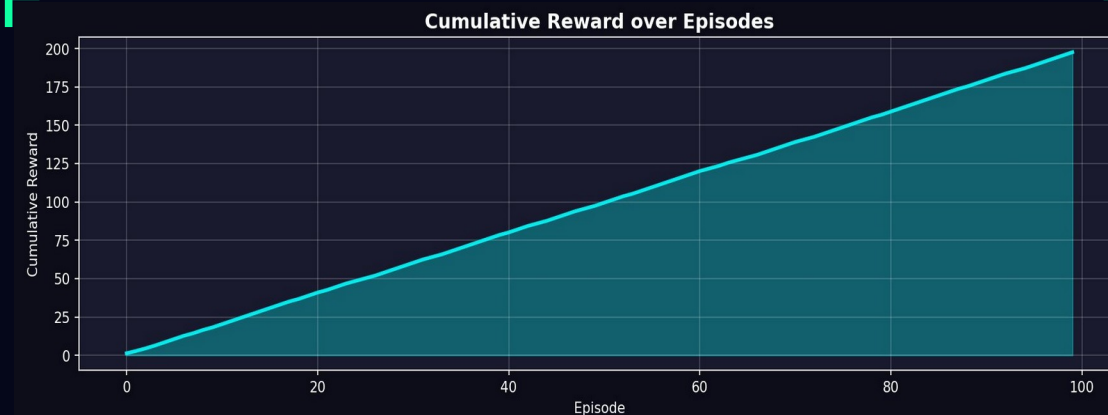


RESULTS · GRPO TRAINING

Qwen 2.5 0.5B · LoRA · 30 steps



TRAINING LOSS + REWARD · 30 STEPS



CUMULATIVE + ROLLING AVG · 100 EPISODES

Real GRPO gradient updates · loss spike at step 23 = high-reward sample · per-step reward 0.1 → 1.5 · proves the env is policy-gradient-friendly

IMPACT + WHAT'S NEXT

Auto-curriculum is just the start

BEYOND CODE

Same blueprint generalizes to math proofs, debugging, theorem reasoning, and any verifiable domain.

TRAINING-READY ENV

GRPO works out of the box. Plug in any small open-source model and start improving it.

FUTURE: MULTI-DOMAIN

Expand difficulty ladders across topics. Let the agent choose its own learning frontier.

TRY IT YOURSELF

HF SPACE huggingface.co/spaces/Vamppog/Auto-examiner

GITHUB github.com/Vamp6969/Auto-examiner

COLAB colab.research.google.com/drive/10okb1w9NMoAgWGt-Ioau8KK0TeEfhATB